

Year 7 • Chapter 5B • Equations: answers (teacher copy)

5.10 Guess, check and improve (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: $p + 8 = 13 \rightarrow p = 5$
- Column 1, E2: $w - 11 = 9 \rightarrow w = 20$
- Column 2, E1: $3w + 1 = 22$; try $w = 7 \rightarrow 3 \times 7 + 1 = 22$; $22 = 22$, so yes
- Column 2, E2: $3p + 4 = 22$; try $p = 7 \rightarrow 3 \times 7 + 4 = 25$; $25 \neq 22$, so no
- Column 3, E1: $3p + 17 = 50 \rightarrow$ try 9: $3 \times 9 + 17 = 44$ (too small); try 12: $3 \times 12 + 17 = 53$ (too big); try 11: $3 \times 11 + 17 = 50 \checkmark$
- Column 3, E2: $3y + 19 = 58 \rightarrow$ try 11: $3 \times 11 + 19 = 52$ (too small); try 14: $3 \times 14 + 19 = 61$ (too big); try 13: $3 \times 13 + 19 = 58 \checkmark$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------|--------------|--------------|
| 1. $t = 12$ | 17. $b = 7$ | 33. no |
| 2. $k = 10$ | 18. $m = 6$ | 34. yes |
| 3. $b = 4$ | 19. $x = 11$ | 35. yes |
| 4. $b = 10$ | 20. $p = 7$ | 36. no |
| 5. $m = 13$ | 21. $y = 10$ | 37. no |
| 6. $a = 3$ | 22. $k = 8$ | 38. yes |
| 7. $w = 12$ | 23. $x = 11$ | 39. yes |
| 8. $w = 13$ | 24. $k = 13$ | 40. yes |
| 9. $k = 6$ | 25. $y = 7$ | 41. $x = 15$ |
| 10. $t = 5$ | 26. $y = 2$ | 42. $y = 7$ |
| 11. $w = 3$ | 27. $t = 22$ | 43. $y = 12$ |
| 12. $n = 25$ | 28. $x = 1$ | 44. $t = 8$ |
| 13. $m = 12$ | 29. $b = 15$ | 45. $n = 12$ |
| 14. $y = 2$ | 30. $n = 9$ | 46. $w = 14$ |
| 15. $w = 7$ | 31. yes | 47. $w = 11$ |
| 16. $b = 13$ | 32. yes | |

5.11 Solving one-step equations (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: $a - 3 = 29 \rightarrow a = 32$
- Column 1, E2: $a - 4 = 24 \rightarrow a = 28$
- Column 2, E1: $9p = 99 \rightarrow 9p \div 9 = 99 \div 9$; $p = 11$
- Column 2, E2: $\frac{k}{7} = 5 \rightarrow \frac{k}{7} \times 7 = 5 \times 7$; $k = 35$
- Column 3, E1: $k + 5 = -9 \rightarrow k + 5 - 5 = -9 - 5$; $k = -14$
- Column 3, E2: $7a = 6 \rightarrow 7a \div 7 = 6 \div 7$; $a = \frac{6}{7}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------|--------------|-------------------------|
| 1. $x = 17$ | 19. $k = 22$ | 37. $x = 12$ |
| 2. $p = 23$ | 20. $y = 30$ | 38. $x = 49$ |
| 3. $t = 11$ | 21. $b = 15$ | 39. $n = 2$ |
| 4. $t = 14$ | 22. $k = 20$ | 40. $x = 22$ |
| 5. $n = 15$ | 23. $a = 17$ | 41. $w = 8$ |
| 6. $w = 14$ | 24. $n = 17$ | 42. $n = 66$ |
| 7. $x = 1$ | 25. $y = 18$ | 43. $w = 12$ |
| 8. $b = 15$ | 26. $b = 33$ | 44. $k = 81$ |
| 9. $x = 4$ | 27. $a = 11$ | 45. $w = 2$ |
| 10. $w = 26$ | 28. $n = 23$ | 46. $b = 12$ |
| 11. $n = 28$ | 29. $p = 21$ | 47. $n = 8$ |
| 12. $t = 12$ | 30. $m = 21$ | 48. $a = 72$ |
| 13. $p = 14$ | 31. $m = 6$ | 49. $x = -1$ |
| 14. $n = 42$ | 32. $a = 14$ | 50. $m = \frac{28}{10}$ |
| 15. $m = 1$ | 33. $t = 4$ | 51. $w = -13$ |
| 16. $w = 22$ | 34. $y = 72$ | 52. $w = \frac{23}{8}$ |
| 17. $t = 34$ | 35. $m = 4$ | 53. $y = -7$ |
| 18. $p = 15$ | 36. $w = 8$ | |

54. $y = \frac{6}{4}$
55. $t = -2$

56. $n = \frac{9}{6}$
57. $x = -11$

5.12 Solving two-step equations (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: $8k - 8 = 51 \rightarrow + 8$
- Column 1, E2: $3p + 16 = 57 \rightarrow - 16$
- Column 2, E1: $2y - 12 = -10 \rightarrow 2y = -10 + 12 = 2$; $y = 2 \div 2 = 1$
- Column 2, E2: $9w - 5 = 31 \rightarrow 9w = 31 + 5 = 36$; $w = 36 \div 9 = 4$
- Column 3, E1: $6(b + 4) = 66 \rightarrow b + 4 = 66 \div 6 = 11$; $b = 11 - 4$; $b = 7$
- Column 3, E2: $9b + 5 = -31 \rightarrow 9b = -31 - 5 = -36$; $b = -36 \div 9$; $b = -4$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------|--------------|--------------|
| 1. $+ 6$ | 21. $- 9$ | 41. $n = 1$ |
| 2. $+ 10$ | 22. $+ 2$ | 42. $x = 2$ |
| 3. $- 8$ | 23. $- 17$ | 43. $n = 2$ |
| 4. $- 2$ | 24. $- 14$ | 44. $b = 8$ |
| 5. $+ 14$ | 25. $+ 17$ | 45. $k = 10$ |
| 6. $- 19$ | 26. $+ 19$ | 46. $n = 6$ |
| 7. $- 18$ | 27. $+ 5$ | 47. $x = 9$ |
| 8. $- 20$ | 28. $+ 1$ | 48. $a = 1$ |
| 9. $+ 20$ | 29. $- 13$ | 49. $t = 6$ |
| 10. $+ 20$ | 30. $p = 6$ | 50. $m = 10$ |
| 11. $- 5$ | 31. $b = 2$ | 51. $w = 2$ |
| 12. $- 2$ | 32. $b = 8$ | 52. $w = 9$ |
| 13. $+ 9$ | 33. $k = 4$ | 53. $m = -5$ |
| 14. $+ 3$ | 34. $p = 12$ | 54. $w = 12$ |
| 15. $+ 3$ | 35. $p = 6$ | 55. $b = -3$ |
| 16. $+ 6$ | 36. $k = 6$ | 56. $m = 3$ |
| 17. $+ 15$ | 37. $p = 9$ | 57. $w = -7$ |
| 18. $+ 17$ | 38. $k = 12$ | 58. $k = 5$ |
| 19. $+ 11$ | 39. $a = 4$ | |
| 20. $- 14$ | 40. $b = 8$ | |

5.13 Equations and problem solving

(booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: A number plus 8 is 14. $\rightarrow n + 8 = 14$
- Column 1, E2: 6 times a number is 108. $\rightarrow 6n = 108$
- Column 2, E1: Double a number, plus 3, is 37. $\rightarrow 2n + 3 = 37$; $n = 17$
- Column 2, E2: Three times a number, minus 6, is 39. $\rightarrow 3n - 6 = 39$; $n = 15$
- Column 3, E1: 10 equal packs hold 200 pencils. How many pencils are in each pack? $\rightarrow 10p = 200$; $p = 200 \div 10 = 20$; There are 20 pencils in each pack.
- Column 3, E2: A plumber charges \$55 call-out plus \$21 an hour. The bill is \$139. How many hours? $\rightarrow 55 + 21h = 139$; $21h = 84$, $h = 4$; The plumber worked 4 hours.
- Column 3, E3: Two angles of a triangle are 26° and 59° . Find the third angle, x° . $\rightarrow 26 + 59 + x = 180$; $x = 180 - 85 = 95$; The third angle is 95° .

Questions (in order: Easy, Medium, Challenging)

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|--------------------|---------------------|--------------------|
| 1. $n - 5 = 4$ | 8. $5n = 50$ | 15. $n + 8 = 23$ |
| 2. $n \div 5 = 10$ | 9. $n - 4 = 19$ | 16. $2n = 8$ |
| 3. $n + 2 = 15$ | 10. $n \div 3 = 6$ | 17. $n - 4 = 8$ |
| 4. $3n = 18$ | 11. $n + 3 = 20$ | 18. $n \div 5 = 8$ |
| 5. $n - 9 = 20$ | 12. $5n = 75$ | 19. $n = 5$ |
| 6. $n \div 5 = 12$ | 13. $n - 5 = 18$ | 20. $n = 18$ |
| 7. $n + 6 = 13$ | 14. $n \div 3 = 20$ | 21. $n = 18$ |

22. $n = 8$
23. $n = 8$

24. $n = 7$
25. $n = 9$

26. $n = 16$
27. $n = 16$

28. 17 pencils
29. 4 hours

30. $x = 83$
31. 11 pencils